Interpretable Prediction of Student Dropout Using Explainable AI Models

Prasadini Padmasiri
Faculty of Graduate Studies and Research
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
prasadinipadmasiri@gmail.com

Sanvitha Kasthuriarachchi
Faculty of Computing
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
sanvitha.k@sliit.lk

***Abstract*—The research addresses the challenge of student dropouts in education institutions, recognizing the pivotal role of education in shaping the society. A research gap is identified, with limited studies effectively combining Machine Learning (ML) algorithms and Explainable AI (XAI) techniques for predicting student dropout. To bridge this gap, a prediction model is developed, integrating ML with XAI to enhance transparency and interpretability. A web interface is introduced, enabling users to input attributes and receive predictions with explanations interpreted using LIME. Further, global interpretation using SHAP is conducted, offering insights into the overall impact and importance of features in the ML model across its dataset. The model is evaluated by selecting academic professionals through purposive sampling technique, employing Likert scale questions, resulting in a mean value exceeding 3.4 and affirming the model's effectiveness in predicting student dropouts with transparency.**

***Index Terms*—dropout prediction, higher education, interpretable machine learning, explainable AI**

I. Introduction

Education is an essential element of modern society, and educational institutions play a significant role in shaping the future of society. However, one of the most significant challenges faced by these institutes is the high rate of student dropouts [1], [2]. The reasons behind student dropout are complex and can vary from personal and financial issues to academic struggles and a lack of motivation [3]. The negative impacts of student dropout are manifold, ranging from individual consequences for the students themselves to economic and social impacts on society as a whole [4], [5]. The issue of student dropout rates has been a significant concern for educators, policymakers, and researchers worldwide. Hence, identifying and addressing these factors in a timely and effective manner is crucial to reducing the dropout rate and ensuring student success.

In the conventional landscape of education, the assessment of student dropout rates has traditionally been a labor-intensive and error-prone process, primarily reliant on manual data collection and analysis [6], [7]. This method involves the painstaking compilation of diverse data points, including academic performance, attendance records, and socio-economic factors, followed by a qualitative analysis to identify potential indicators of student attrition. However, this approach is not without its limitations; it often proves time-consuming, subject to human error, and may not provide real-time insights into the dynamic factors influencing student retention.

979-8-3503-7568-8/24$31.00 ©2024 IEEE

The advent of big data and advancements made in machine learning technologies have placed educational institutions in a revolutionary position [8], [9]. The abundance of data generated by educational institutions, ranging from academic records and student demographics to engagement patterns in online learning environments, presents a rich source for analysis. Machine learning algorithms, equipped with the capability to discern patterns and relationships within vast datasets, offer a paradigm shift in how institutions can approach the complex challenge of predicting student dropout rates.

In dropout prediction studies, machine learning algorithms are commonly employed for accurate classification. However, the interpretation of these models and explanations for predictions is often neglected [9]. Beyond achieving high classification accuracy, effective decision support systems should provide insights into why predictions are made, suggesting personalized interventions for individuals. This not only enables tailored guidance and support for students but also enhances model transparency, establishing trust with decision-makers. To address this, interpretable machine learning (IML) or explainable artificial intelligence (XAI) tools are utilized to demystify the workings of black-box models.

This research is primarily geared towards utilizing predictive modeling and XAI to advance student retention in higher education. The main goal involves crafting a predictive model proficient in accurately determining whether a student is at risk of dropping out. This model considers a spectrum of factors, including academic performance and demographic information, to provide a comprehensive evaluation of the student's situation. Upon completion of this research, the following Research Questions (RQs) had been answered:

- RQ1: 1. What are the key factors contributing significantly to student dropout in higher education, as identified through the analysis of the dataset using XAI?
- RQ2: 2. How can a model be effectively implemented to achieve precise identification of student dropout or retention while concurrently providing comprehensive explanations for each prediction?

The ultimate output of the research is twofold: firstly, the prediction of whether a student is likely to become a dropout, and secondly, the provision of transparent explanations for this prediction through XAI. As the model identifies students facing potential challenges, the ensuing interventions will be tailored based on a precise understanding of the underlying reasons, ensuring a personalized and effective approach. In summary, this research aims to deliver a predictive model capable of discerning dropout risks and providing comprehensible insights through XAI, thereby contributing to the proactive enhancement of student retention in higher education.

II. Related Work

The purpose of this literature review is to provide an overview of current research focused on the development of machine learning models for predicting university student dropout rates. To prevent confusion, it is crucial to define the term "university dropout" precisely. A study [3] examined different interpretations of the term and concluded that it refers to the failure to complete a specific university program. It is important to distinguish the point at which dropout occurs because students may switch majors but continue studying in the same field, transfer to a different university for various reasons, or completely leave the university system [10]. Student dropout has been a long-standing issue in educational institutions, and researchers have conducted numerous studies to identify contributing factors and develop effective prediction and prevention methods.

A. Identification of Factors Contributing to Student Dropout

Regarding the identification of factors contributing to student dropout, one study [5] compared dropout rates and causes in two different distance education courses. The authors identified several factors that contributed to high dropout rates, including a lack of interaction with instructors and peers, difficulty in navigating course materials, and a lack of motivation. A study [11] investigated the effects of extracurricular activities, ethnic identification, and school perception on student dropout rates, focusing on Latino students who have a higher dropout rate compared to other ethnic groups. Participating in extracurricular activities was found to be negatively associated with dropout rates, suggesting that involvement in school activities can promote student engagement and retention. Additionally, a positive perception of school and a strong ethnic identity were associated with lower dropout rates.

Additionally, a study [12] investigated the factors contributing to engineering student dropout in distance learning. Statistical analysis and machine learning algorithms were utilized to identify factors associated with student dropout. The study revealed that factors such as low grades, low participation in discussion forums, and low quiz scores significantly increased the likelihood of dropout. The authors emphasized that data-driven approaches can assist educators in developing targeted interventions to prevent student dropout.

Furthermore, a paper [13] focused on identifying the optimal time and data for predicting undergraduate student dropout rates in universities. The study involved a retrospective analysis of student data from a large Spanish university and identified key variables influencing student retention rates, including student demographics, academic performance, and socioeconomic factors. The results indicated that the first semester was the most opportune time to identify at-risk students, and academic performance and demographics were the most significant variables for predicting student dropout rates.

B. Predictive Models using Machine Learning Algorithms

In recent years, the utilization of machine learning algorithms to predict and mitigate the challenges of university dropout rates has emerged as a pivotal area of research. This section explores the burgeoning field of predictive models for anticipating university dropout, specifically focusing on the application of machine learning algorithms. The advent of sophisticated algorithms, including logistic regression, decision trees, and other machine learning techniques, has provided researchers with powerful tools to analyze complex datasets and discern patterns indicative of potential student dropout.

Drawing on data from an international research project, the study analyzes Learning Management System (LMS) tracking data from a Blackboard Vista-supported course [14]. It identifies 15 variables with a significant correlation to students' final grades. Through regression modeling, a predictive model is developed, incorporating key variables such as the total number of discussion messages, mail messages, and assessments completed. In reference [15], the authors employ academic data, test results, and participation in activities gathered from the LMS of a Midwestern U.S. University. Employing a feature selection method for enhanced generalizability, the Naive Bayes Classifier and an Ensemble model incorporating various methods showed the most promising results.

The study [16] focuses on utilizing students' personal characteristics and academic performance as input features for prediction models. Artificial Neural Network (ANN), Decision Tree (DT), and Bayesian Networks (BNs) were employed to develop these models. Another study [17] aimed at predicting student dropout at the Karlsruhe Institute of Technology (KIT), have employed two machine learning approaches: logistic regression and decision trees. The analysis incorporates a few biographical parameters that the university routinely collects during the course enrollment process and detailed observations of 487 different examinations, encompassing information on taken examinations, grades, dates, results, and the number of attempts. Moreover, a paper [18] explored the use of machine learning algorithms to predict high school dropout rates beyond early warning indicators. Machine learning algorithms like random forest and gradient boosting outperformed traditional early warning indicators in predicting high school dropout rates. Several risk factors, including attendance, be-

havior, and academic performance, were strongly associated with high school dropout rates.

C. Predictive Models using Explainable AI (XAI)

XAI serves as a navigational tool, unraveling the intricacies of decision-making within sophisticated computer models. It acts as an insight into the hidden workings of algorithms, clarifying their predictions for enhanced transparency [19]. This section delves into the evolving realm of predictive models using XAI, aiming to shed light on how XAI techniques enhance the interpretability of these models, particularly in the context of identifying at-risk students.

The research paper [19] discusses the development of an XAI predictive model that can identify students at risk of failure in online courses. The model analyzes students' learning behavior, such as online forum participation, interaction with the instructor, repetition rate, problem-solving rate, and quiz attempts. The model provides personalized feedback and assistance to students and instructors in decision-making. The study [20] proposes an intelligent framework for explainable student performance prediction using two innovative prediction models called CNN-LSTM and Conv-LSTM. The study aims to improve the user experience during the learning process by providing detailed-class labeling and enhancing the school environment by supporting decision-makers in implementing early intervention techniques to reduce the risk of students falling behind.

The study [9] explores the use of XAI tools to predict student dropout rates in higher education. The paper compares the output and readability of different XAI tools and presents a user study where students and higher education decision-makers were surveyed about the applied tools. The study assessed the participants' data visualization literacy and asked their opinion about the interpretability, utility, and design of the charts. Another study [21] explores on the role of AI/ML in the education domain, with a focus on student performance prediction. The study highlights the need for interpretable model outcomes and explains the factors that make the results interpretable. The study reviews EDM and the various algorithms used for prediction, as well as the gaps in the field.

Overall, these studies highlight the importance of comprehending the factors contributing to student dropout and developing effective methods for prediction and prevention. Machine learning approaches have shown promise in predicting student dropout, but further research is required to explore additional factors and interventions.

III. Methodology

The methodology, comprising several strategic steps, involves acquiring a pertinent dataset from Kaggle, preprocessing the data, deploying diverse predictive modeling algorithms, evaluating model performance rigorously, integrating XAI techniques for transparency, and seeking validation from academic professionals in higher education. The high-level architecture depicted in Fig. 1 provides an overview of the prediction system.

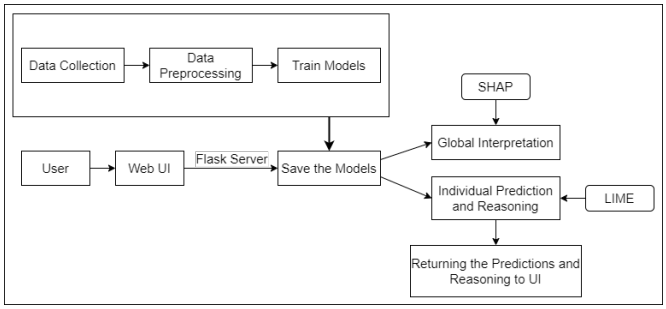


Fig. 1. High-level architectural diagram

A. Data Collection

This dataset sourced from Kaggle provides a comprehensive overview of students enrolled in various undergraduate programs at a higher education institution. It covers a wide range of information, including demographic details, socio-economic factors, and academic performance metrics. The dataset can be accessed at [22]. It is composed of multiple databases that contain relevant information from the time of enrollment, such as application mode, marital status, chosen course, and more. Additionally, the data facilitates the assessment of overall student performance at the conclusion of each semester, taking into account factors like credited curricular units, enrollment status, evaluation outcomes, approvals, and corresponding grades. Moreover, the dataset includes regional indicators like the unemployment rate, inflation rate, and GDP, offering insights into how economic factors may impact student dropout rates or academic success.

Analyzing the distribution of outcomes in the 'Target' column, which categorizes students into 'Dropout,' 'Enrolled,' and 'Graduate,' reveals insightful patterns. The dataset consists of 49.93% graduates, 32.12% dropouts, and 17.94% enrolled students in the 'Target' column, reflecting the distribution of outcomes among the student population.

B. Data Preprocessing

Data preprocessing, a facet of data preparation, refers to any form of processing applied to raw data to ready it for subsequent data processing procedures. The phase involved several key steps, including the renaming of column names, handling null values, encoding categorical variables, and implementing feature engineering.

1) Renaming Column Names: Column names were scrutinized to ensure clarity and relevance. This involved eliminating ambiguities, using standardized naming conventions, and aligning labels with the specific research context, creating a more intuitive and informative dataset. Here, two columns underwent notable adjustments. The label "Nacionality" was refined to "Nationality" and the column labeled "Age at enrollment" was succinctly renamed to "Age".

2) Handling Null Values: A comprehensive analysis of null values was conducted, identifying the extent and pattern of missing data. In the examination for null values during the data preprocessing phase, a comprehensive check was conducted, specifically focusing on potential missing data within the dataset. Fortunately, this scrutiny revealed no instances of null values.

3) Encoding Categorical Variables: The encoding of the target variable, representing distinct categories in the context of student outcomes, has been addressed. Initially, the target variable encompassed three categories: "Dropout" "Enrolled" and "Graduate" and a systematic encoding approach was adopted. Due to the limited number of unique values in the target variable, a practical approach was adopted for encoding using the map() method. This method is particularly effective when dealing with a small set of distinct categories. Accordingly, the categories were replaced with numerical values, where "Dropout" was encoded as 0, "Enrolled" as 1, and "Graduate" as 2.

4) Feature Engineering: Feature engineering involved the creation of new variables to enhance the dataset's predictive power. A thorough examination was conducted to understand the correlation between features and the target variable. This involved assessing how each feature relates to the target variable, providing insights into their individual contributions to the prediction task. Features exhibiting strong correlations with the target were retained, while those with weaker associations were carefully considered for potential removal or transformation.

Another significant aspect of feature engineering involved the identification and removal of features that were deemed irrelevant or extraneous to the predictive task. Unwanted features, which may introduce noise or unnecessary complexity, were systematically eliminated. The dataset underwent a correlation analysis using a heatmap, revealing columns with low absolute correlation coefficients with the target variable ('Target'). To prioritize more influential variables, a predefined correlation threshold of 0.1 (absolute value) was established. Columns displaying correlations below this threshold were identified for potential removal. As a result, the following columns were removed from the dataset: Nationality, Mother's Qualification, Father's Qualification, Educational Special Needs, International, Curricular Units 1st Sem (Without Evaluations), Unemployment Rate, and Inflation Rate.

C. Machine Learning Approach

For the development of the model, classification algorithms were employed, given the discrete nature of the target variable. The dataset was divided into training and test sets using a random splitting method, with 20% of the data allocated to the test set to ensure robust evaluation. The dataset was trained multiple times to allow the model to be trained and evaluated iteratively, facilitating the identification and prevention of overfitting.

The selected algorithms for model development include Decision Tree, Random Forest Classifier, K-Neighbors Classifier, AdaBoost Classifier, and Support Vector Machine (SVM). The performance of the classifiers was assessed using key evaluation metrics including accuracy, precision, recall, and F1 score. Impressively, all evaluation scores for the classification models fall within the range of 0.69 to 1.0. Notably, both the Random Forest Classifier and Decision Tree stand out as top performers, achieving a score of 1.0 on the training set and 0.69 to 0.8 on the test set. The detailed results of the evaluation metrics are presented in Fig. 2.

D. Explainable AI Models

Explainable AI models were strategically selected to illuminate the decision-making process of the model. Among these, the SHAP (SHapley Additive exPlanations) technique was chosen for its robustness and capability to offer a global interpretation for the dataset [23], [24]. In contrast to other methods, SHAP allows for the aggregation of information from multiple local explanations, enabling the identification of common trends and patterns exhibited by the model across the entire dataset. This approach facilitates a deeper understanding of which features consistently contribute to the model's predictions across various instances and elucidates overarching trends in feature importance. Furthermore, the utilization of LIME (Local Interpretable Model-agnostic Explanations) provided additional insights into individual predictions, offering local interpretability and aiding in understanding the model's behavior [23], [25]. While LIME excels in providing interpretable insights at the local level, SHAP offers a more comprehensive view of the model's decision-making process at a global level, thus justifying its preference in this study.

E. Development of the Web Application

During the interactive phase of implementation, characterized by a crucial emphasis on user engagement, a React web application functions as the interface. Within this framework, users are afforded the opportunity to input specific attributes into the application and subsequently submit their requests. A Flask server assumes responsibility for managing these requests, accessing the stored model. The output encompasses predictions, accompanied by interpreted rationales derived through the application of LIME.

LIME adopts a perturbation-based approach, creating a perturbed dataset to train an explainable model [26]. In contrast, SHAP relies on calculating SHAP values across an entire sample, necessitating multiple observations. The fundamental difference lies in the sample size requirement—LIME can effectively operate with just one observation, making it particularly suited for scenarios where limited data points are available [26]. Given this characteristic, the LIME technique was chosen for the application, aligning with the web-based prediction tool that processes user inputs individually to provide concise and insightful local explanations for each prediction.

	Train_Accuracy Score	Train_Precision Score	Train_F1Score	Train_Recall	Test_Accuracy Score	Test_Precision Score	Test_F1Score	Test_Recall	Test_AUC_Score
Random Foreset Classifier	1.000000	1.000000	1.000000	1.000000	0.805650	0.805650	0.805650	0.805650	0.904037
AdaBoost Classifier	0.767731	0.767731	0.767731	0.767731	0.771751	0.771751	0.771751	0.771751	0.851887
SVM	0.773947	0.773947	0.773947	0.773947	0.770621	0.770621	0.770621	0.770621	0.886797
Decision Tree	1.000000	1.000000	1.000000	1.000000	0.693785	0.693785	0.693785	0.693785	0.734482
K-Neighbors Classifier	0.822831	0.822831	0.822831	0.822831	0.693785	0.693785	0.693785	0.693785	0.783356

Fig. 2. Results of evaluation metrics

F. Evaluation Process

Evaluation serves as the conclusive assessment of the proposed approach, offering a comprehensive examination and insights to enhance the research. To capture a range of perspectives from academic professionals, a purposive sampling technique was used, with 15 participants selected. The non-probability sampling method of purposive sampling was employed, relying on the judgment to choose elements for the sample [27]. The selection criteria focused on the academic experience of the respondents.

IV. Results

This section presents a summary of the results obtained from the research and mainly two RQs stated in *Section I: Introduction* are primarily addressed.

Global interpretation with SHAP technique involves analyzing the overall impact and importance of features in a machine learning model across its entire dataset [28]. By assigning Shapley values to each feature, this approach provides a comprehensive understanding of how individual variables contribute to model predictions on a global scale, fostering transparency and interpretability in complex models [29]. In this research, the global interpretation using the SHAP technique was conducted on two prominent machine learning algorithms: the Random Forest Classifier and Decision Tree.

Fig. 3 displays the SHAP summary plot for the Random Forest Classifier. The plot visually represents the distribution of SHAP values across different features, offering insights into how each feature contributes to the model's predictions on a global scale. Fig. 4 showcases the SHAP summary plot for the Decision Tree algorithm. Similar to the Random Forest plot, it presents the distribution of SHAP values for different features within the Decision Tree model.

The first RQ explores the key factors contributing significantly to student dropout in higher education. According to the global interpretation with the machine learning algorithms, a notable emphasis emerges on the influence of specific attributes. Notably, both plots, as illustrated in Fig. 3 and Fig. 4 provides a concise yet insightful depiction of how key variables, such as curricular units 2nd sem (approved), curricular

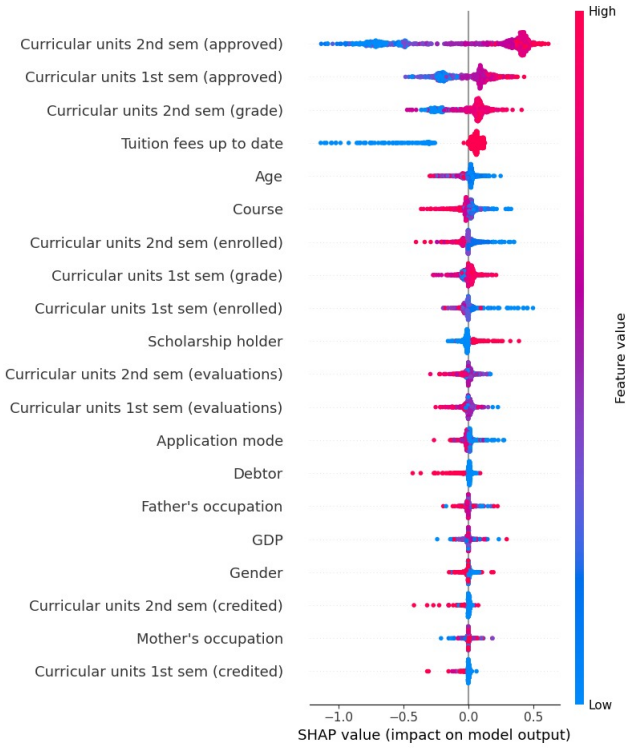


Fig. 3. Global interpretation by SHAP with Random Forest Classifier

units 1st sem (approved), and tuition up to date, influence the model's prediction of student dropout. Higher values of these variables exert a positive impact on the prediction, indicating a reduced likelihood of dropout, while lower values have a negative impact, signifying an increased risk of dropout. Upon deeper analysis of the summary plots, it becomes evident that academic records, financial status, age, and the chosen course stand out as pivotal factors significantly influencing the model predictions. These variables play a crucial role in shaping the model's outlook on the likelihood of student dropout.

Answering to the second RQ, a web application has been developed to facilitate user to input attributes and receive predictions, with explanations interpreted using LIME. The explanations behind the model predictions are depicted through visual plots. A key objective of XAI tools is to generate results that are readily understandable by end users, fostering

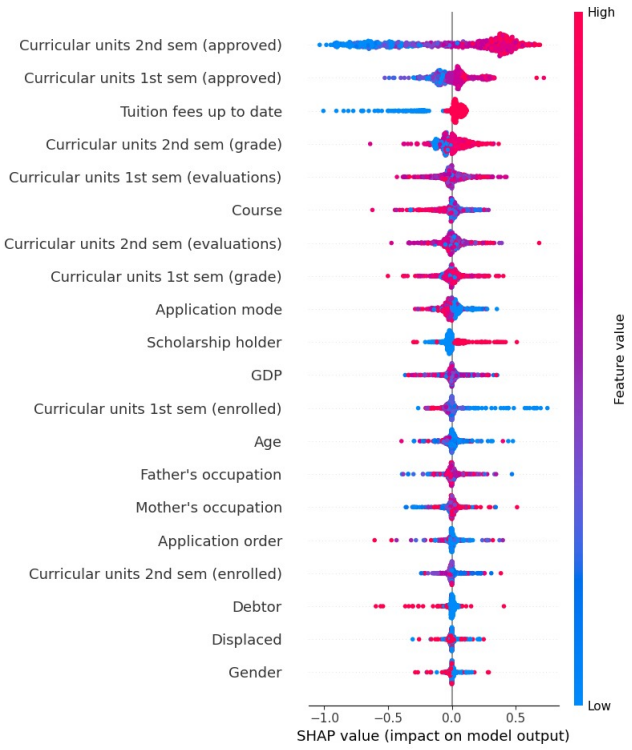


Fig. 4. Global interpretation by SHAP with Decision Tree

trust among decision-makers and stakeholders [9]. Despite the recent introduction of various Interpretability Machine Learning (IML) methods, there is limited research on the usability and transparency of these tools, particularly regarding their interpretability at the application level [9].

The criterion employed for evaluation, along with their respective descriptions, are mentioned in TABLE I.

TABLE I. EVALUATION CRITERION, DESCRIPTION AND MEAN SCORES

Criterion	Description	Mean
Necessity of the application	Evaluates the industry's perceived need for and acceptance of the tool	3.53
Satisfaction on given predictions	Measures user satisfaction with the application's predictions, focusing on relevance and effectiveness	3.73
Accuracy of the application	Evaluates the application's overall precision	3.6
Comprehensibility of the explanations	Assess how well users understand the local explanations	3.4
Impact on decision making	Assess the practical impact of the predictions and explanations on users' decision-making processes	3.93

Considering the respondents' (n=15), demographic information, the average length of respondents' total years of academic experience is 3.7 years. The perceived necessity of the application garnered a mean score of 3.53, suggesting a moderate level of acknowledgment for its relevance. Users expressed satisfaction with the provided predictions, yielding a mean score of 3.73, signifying a favorable response to the accuracy and relevance of the model's outputs. The application's overall accuracy received a commendable mean score of 3.6, reflecting a perceived effectiveness in generating precise predictions. While the comprehensibility of explanations earned a mean score of 3.4, indicating a moderate level of clarity, the application's notable strength lies in its impact on decision-making processes, with a high mean score of 3.93. This suggests that users find the predictions influential in guiding their decisions, emphasizing the practical utility of the XAI-driven prediction model.

V. Conclusion

In conclusion, this research has successfully tackled a prominent challenge in higher education by presenting a dropout prediction model that integrates machine learning with Explainable AI techniques. The study filled a noteworthy research gap, demonstrating the efficacy of combining these approaches to enhance transparency and interpretability in predicting student dropouts. The user-friendly React web application, paired with the Flask server, illustrates a practical solution during the interactive phase of implementation. Users, empowered by the simplicity of the interface, can input attributes and receive predictions. The integration of LIME provides individualized explanations, opening avenues for personalized interventions and shedding light on the subtle explanations behind each prediction.

By addressing the identified research gap, this work contributes significantly to the academic discourse on student attrition in higher education. The model's practical implications are notable, suggesting its applicability in real-world educational settings for identifying and supporting students at risk of dropout. The user-centric approach, emphasizing the interpretability of predictions, adds a valuable dimension to understanding dropout risk factors and facilitating targeted interventions.

Insights from the global interpretation using SHAP further enrich the overall understanding of feature importance in the machine learning model. This dual approach not only provides transparent and accurate predictions but also equips decision-makers with valuable information for informed educational strategies.

Looking forward, potential future directions include refining the model, exploring additional features, and expanding its application to different educational contexts. In essence, this research underscores the transformative potential of integrating machine learning and Explainable AI in addressing the complex issue of student dropouts, offering a comprehensive and user-focused approach for educational institutions.

References

[1] M. Segura, J. Mello, and A. Hernández, "Machine learning prediction of university student dropout: Does preference play a key role?" *Mathematics*, vol. 10, p. 3359, 9 2022.

[2] M. Solis, T. Moreira, R. Gonzalez, T. Fernandez, and M. Hernandez, "Perspectives to predict dropout in university students with machine learning." IEEE, 7 2018, pp. 1–6.

[3] M. S. Larsen, K. P. Kornbeck, R. M. Kristensen, M. R. Larsen, and H. B. Sommersel, *Dropout phenomena at universities : what is dropout? Why does dropout occur? What can be done by the universities to prevent or reduce it? : a systematic review*. Danish Clearinghouse for Educational Research, 2013.

[4] F. Araque, C. Roldán, and A. Salguero, "Factors influencing university drop out rates," *Computers & Education*, vol. 53, pp. 563–574, 11 2009.

[5] C. Pierrakeas, M. Xeno, C. Panagiotakopoulos, and D. Vergidis, "A comparative study of dropout rates and causes for two different distance education courses," *The International Review of Research in Open and Distributed Learning*, vol. 5, 8 2004.

[6] V. Tinto, "Dropout from higher education: A theoretical synthesis of recent research," *Review of Educational Research*, vol. 45, pp. 89–125, 3 1975.

[7] W. Xing and D. Du, "Dropout prediction in moocs: Using deep learning for personalized intervention," *Journal of Educational Computing Research*, vol. 57, p. 073563311875701, 03 2018.

[8] A. J. Fernandez-Garcia, J. C. Preciado, F. Melchor, R. Rodriguez-Echeverria, J. M. Conejero, and F. Sanchez-Figueroa, "A real-life machine learning experience for predicting university dropout at different stages using academic data," *IEEE Access*, vol. 9, pp. 133 076–133 090, 2021.

[9] M. Nagy and R. Molontay, "Interpretable dropout prediction: Towards xai-based personalized intervention," *International Journal of Artificial Intelligence in Education*, 3 2023.

[10] "Dropping out of university: a literature review," *Review of Education*, vol. 8, pp. 614–652, 6 2020.

[11] D. B. Davalos, E. L. Chavez, and R. J. Guardiola, "The effects of extracurricular activity, ethnic identification, and perception of school on student dropout rates," *Hispanic Journal of Behavioral Sciences*, vol. 21, pp. 61–77, 2 1999.

[12] S. Rovira, E. Puertas, and L. Igual, "Data-driven system to predict academic grades and dropout," *PLOS ONE*, vol. 12, p. e0171207, 2 2017.

[13] J. M. Ortiz-Lozano, A. Rua-Vieites, P. Bilbao-Calabuig, and M. Casadesús-Fa, "University student retention: Best time and data to identify undergraduate students at risk of dropout," *Innovations in Education and Teaching International*, pp. 1–12, 8 2018.

[14] L. P. Macfadyen and S. Dawson, "Mining lms data to develop an "early warning system" for educators: A proof of concept," *Computers & Education*, vol. 54, pp. 588–599, 2 2010.

[15] F. Marbouti, H. A. Diefes-Dux, and K. Madhavan, "Models for early prediction of at-risk students in a course using standards-based grading," *Computers & Education*, vol. 103, pp. 1–15, 12 2016.

[16] M. Tan and P. Shao, "Prediction of student dropout in e-learning program through the use of machine learning method," *International Journal of Emerging Technologies in Learning (iJET)*, vol. 10, p. 11, 2 2015.

[17] L. Kemper, G. Vorhoff, and B. U. Wigger, "Predicting student dropout: A machine learning approach," *European Journal of Higher Education*, vol. 10, pp. 28–47, 1 2020.

[18] D. Sansone, "Beyond early warning indicators: High school dropout and machine learning," *Oxford Bulletin of Economics and Statistics*, vol. 81, pp. 456–485, 4 2019.

[19] A. B. Arrieta, N. Díaz-Rodríguez, J. D. Ser, A. Bennetot, S. Tabik, A. Barbado, S. Garcia, S. Gil-Lopez, D. Molina, R. Benjamins, R. Chatila, and F. Herrera, "Explainable artificial intelligence (xai): Concepts, taxonomies, opportunities and challenges toward responsible ai," *Information Fusion*, vol. 58, pp. 82–115, 6 2020.

[20] "Week-wise student performance early prediction in virtual learning environment using a deep explainable artificial intelligence," *Applied Sciences*, vol. 12, p. 1885, 2 2022.

[21] M. Chitti, P. Chitti, and M. Jayabalan, "Need for interpretable student performance prediction." IEEE, 12 2020, pp. 269–272.

[22] (2024) Kaggle. [Online]. Available: https://www.kaggle.com/datasets/thedevastator/higher-education-predictors-of-student-retention

[23] C. Molnar, "Interpretable machine learning a guide for making black box models explainable," 2019. [Online]. Available: http://leanpub.com/interpretable-machine-learning

[24] S. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," 5 2017.

[25] M. T. Ribeiro, S. Singh, and C. Guestrin, ""why should i trust you?": Explaining the predictions of any classifier," 2 2016.

[26] M. Michael and P. David, *Explainable AI for Practitioners: Designing and Implementing Explainable ML Solutions*. O'Reilly, 2022.

[27] (2023) Business research methodology. [Online]. Available: https://research-methodology.net/sampling-in-primary-data-collection/purposive-sampling/

[28] R. Saleem, B. Yuan, F. Kurugollu, A. Anjum, and L. Liu, "Explaining deep neural networks: A survey on the global interpretation methods," *Neurocomputing*, vol. 513, pp. 165–180, 11 2022.

[29] (2023) Datacamp. [Online]. Available: https://www.datacamp.com/tutorial/introduction-to-shap-values-machine-learning-interpretability